

AUDITING MEMORY PROVENANCE IN SELF-EVOLVING AGENTS: CONTROLLED EVIDENCE AND IDENTIFICATION LIMITS

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ABSTRACT

Self-evolving agents convert previous trajectories into reusable textual memory. We study a variable carried into that state that content-only audits can miss: the process provenance of the trajectory from which memory was written. A recent financial-agent audit reports 0.931 accuracy over 101 success-derived ReasoningBank retrievals versus 0.647 over 34 failure-derived retrievals, with all ten reported persistent $W \rightarrow W$ failures in the failure-derived stratum. We introduce a provenance-audit identification ladder that separates observational provenance signals, controlled writer-mode contrasts, exact-information metadata interventions, and source-faithful transport. On released WebArena evidence, R4 evaluates terminal success over four task pairs that pass a frozen pre-outcome semantic-equivalence rule and 48 completed requests. The observed success-derived minus failure-derived difference is +0.1667, but the frozen permutation test gives $p = 0.0785$; with four independent tasks, the exact one-sided sign-flip floor is $1/16 = 0.0625$, so the endpoint is structurally non-decisive under the frozen 0.05 rule. R6 uses 23 disjoint task pairs and 276 completed requests. Its observed reference-action-match difference is -0.0942 with counterevidence-side $p = 0.0792$; paired next actions differ in 21.0% of seeds, but this readout is not terminal quality. R5 targets the cleaner exact-information rung by holding actionable memory byte-identical and changing only provenance metadata, but its frozen ten-task design has only five eligible unique tasks and stops before any model call. Thus the strong observational association survives as an audit signal while the provenance-only causal sign remains unresolved. The reusable contribution is an explicit identification protocol that reports the intervention channel, independent inference unit, assumptions, and strongest allowed claim before provenance is used in memory trust decisions.

1 INTRODUCTION

Self-evolving agents increasingly learn through external memory. A successful episode can become a reusable workflow; a failed episode can become a corrective reflection. Systems such as ReasoningBank convert trajectories into reasoning memories that influence future behavior (Ouyang et al., 2026), while Agent Workflow Memory and Reflexion persist workflows or verbal feedback across trials (Wang et al., 2024; Shinn et al., 2023).

The usual abstraction treats retrieved content as the state variable that matters. Content-centric systems such as A-MEM and MemRL improve memory organization, retrieval, or runtime utility (Xu et al., 2025; Zhang et al., 2026). A second literature makes provenance explicit: recent work organizes execution provenance and evidence tracing for LLM agents (Wang et al., 2026), while Memory Provenance Laundering shows that consolidation can obscure low-trust source provenance while preserving an action trigger (Xu et al., 2026). We therefore do not claim that provenance-aware memory is itself new. Our narrower object is *process provenance*: whether persistent guidance was produced from a successful or failed trajectory, and which causal channel is actually changed when that provenance is manipulated.

Failure provenance also has no justified negative sign a priori. Learning from Failure converts failed computer-use trajectories into diagnoses and patches that improve later behavior (Sun et al., 2026), and controlled multi-round skill-evolution experiments report useful skills in settings that include failure feedback (Liu et al., 2026). These results rule out the shortcut “failure-derived means harmful.” A valid audit must separate an observational provenance association from the intervention used to test it.

A recent audit of self-evolving financial agents supplies the motivating signal (Li & Zhu, 2026). In its ReasoningBank analysis, success-derived retrievals have reported accuracy 0.931 over 101 cases and failure-derived retrievals 0.647 over 34 cases. The audit also reports ten persistent $W \rightarrow W$ failures, all in the failure-derived retrieval stratum. This is a strong association, but task difficulty is an obvious confound: hard tasks can be more likely to fail during acquisition and remain difficult later. A second ambiguity is intervention design. Switching ReasoningBank’s success- versus failure-reflection objective changes an outcome-conditioned writer; it is not the same operation as holding memory text fixed and changing only provenance metadata.

We formalize these distinctions as a *provenance-audit identification ladder*. L0 is observational stratification by provenance. L1 changes writer mode while fixing the future query/state and enforcing a preregistered semantic-equivalence rule on the generated memories. L2 holds actionable memory byte-identical and changes only visible provenance metadata. L3 transports an identified contrast back to the motivating source runtime. Each rung names the intervention, independent inference unit, identifying assumptions, and strongest allowed conclusion. This makes the protocol applicable beyond ReasoningBank to memory systems in which history changes a writer, a retrieval rule, or explicit metadata.

Our controlled evidence populates the ladder without manufacturing a causal sign. R4 instantiates L1 with four independent terminal task units and 48 completed requests: the observed success-derived minus failure-derived terminal-success difference is $+0.1667$, but $p = 0.0785$ and four units make a one-sided exact $p < 0.05$ unattainable. R6 supplies 23 disjoint L1 task units and 276 completed requests: paired actions differ in 21.0% of seeds, while the observed reference-action-match difference is -0.0942 with counterevidence-side $p = 0.0792$; reference-action match is not terminal quality. R5 targets L2 with byte-identical actionable memory, but the frozen ten-task design exposes only five eligible unique tasks and stops before any model call. Source-faithful L3 replication remains support debt. C4 therefore remains active without a supported provenance-only causal sign; C5 remains a design implication rather than a demonstrated governance benefit.

Contributions and closest-work boundary. We contribute (i) a source-bound process-provenance association from the financial-agent audit, (ii) an L0–L3 identification ladder that distinguishes writer-mode, metadata-only, and transport claims, and (iii) executed L1 endpoints plus an L2 support stop that make the unresolved boundary explicit instead of collapsing all evidence into a single “provenance effect.” We do not claim the first provenance-aware memory system, that failure-derived memories are generally harmful, or that provenance gating improves downstream utility. Relative to provenance tracing and provenance-laundering work, our contribution is the intervention/estimand boundary for success- versus failure-conditioned memory construction; relative to learning-from-failure work, the contribution is an audit protocol with no assumed sign.

2 PROVENANCE AUDIT AND IDENTIFICATION TARGETS

Consider an acquisition episode with task x , trajectory τ , and terminal outcome $z \in \{0, 1\}$. Let $P \in \{S, F\}$ denote whether the source trajectory succeeded or failed. In a provenance-conditioned memory system, P can affect at least two downstream objects: the writer mode $W(P)$ used to construct memory and, when exposed to the agent, a provenance label or metadata field $L(P)$. The written memory is

$$M = G(\tau, W(P)), \quad (1)$$

and future behavior follows

$$Y = H(x', M, L, S), \quad (2)$$

where x' is the future query and S is the remaining agent state. ReasoningBank instantiates the writer channel directly: successful trajectories are analyzed under a success-reflection objective, while failed

Failure provenance is an upstream intervention on memory construction

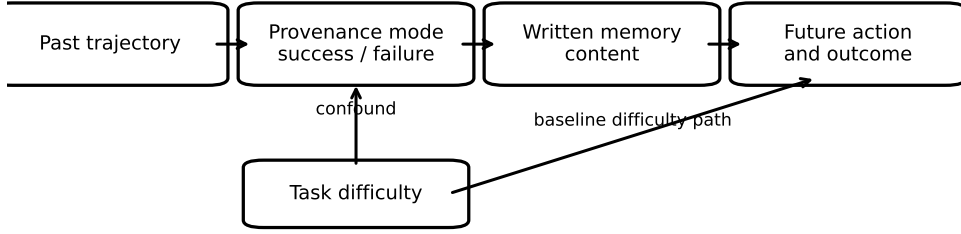


Figure 1: Coarse causal structure of provenance-conditioned memory. Acquisition difficulty can affect both source outcome and later task performance. Writer mode mediates the path from source provenance to written memory; an optional visible provenance-metadata channel is omitted for readability. Matching future queries and semantic guidance reduces the task-difficulty confound but does not make generated memories identical.

trajectories are processed under a failure-reflection objective (Ouyang et al., 2026). A system may also attach an explicit provenance field to the resulting memory.

2.1 A PROVENANCE EFFECT IS NOT A SINGLE ESTIMAND

The observational quantity

$$A_P = \mathbb{E}[Y \mid P = S] - \mathbb{E}[Y \mid P = F] \quad (3)$$

is useful as an audit signal, but it is not causal: acquisition difficulty D can influence both P and future outcomes. More importantly, a notation such as $do(P)$ is underspecified once provenance controls memory construction. One must state which downstream channel is intervened on.

The *writer-mode contrast* changes the reflection objective while fixing the future query and released observation state:

$$\Delta_W(x') = \mathbb{E}[Y \mid do(W = W_F), x', E, \mathcal{G} = 1] - \mathbb{E}[Y \mid do(W = W_S), x', E, \mathcal{G} = 1], \quad (4)$$

where E denotes the frozen future evidence/state and $\mathcal{G}$ is a preregistered information-equivalence gate on the two generated memories. This is the object approximated by our R4 and R6 bridges. Even when $\mathcal{G} = 1$, the intervention can change wording, emphasis, decomposition, or other decision-relevant properties of M ; therefore Δ_W is a writer-mode-bundle estimand, not automatically a provenance-metadata-only effect.

A stricter *exact-information metadata contrast* instead holds the actionable memory text m byte-identical and changes only visible provenance metadata:

$$\Delta_L(m, x') = \mathbb{E}[Y \mid M = m, do(L = F), x', E] - \mathbb{E}[Y \mid M = m, do(L = S), x', E]. \quad (5)$$

This is the cleanest provenance-only object when the deployment system exposes provenance metadata to the policy. Our preregistered R5 design targets this rung, but its support gate fails before model execution because the frozen cohort contract supplies only five unique tasks for a ten-task target.

2.2 PROVENANCE-AUDIT IDENTIFICATION LADDER

Table 1 turns these distinctions into a reusable audit protocol. The key reporting rule is that request count is not automatically the inference unit: R4 contains 48 completed requests but only four independent task units; R6 contains 276 requests but 23 independent task units.

Table 1: Provenance-audit identification ladder.

Rung	Intervention / hold fixed	Current evidence and independent unit	Strongest allowed claim
L0	Observe provenance strata; no intervention	Financial audit; source/retrieval cases	Association / audit signal only
L1	Switch writer mode; fix future query/state; require $G = 1$	R4: 4 task pairs; R6: 23 task pairs	Writer-mode-bundle contrast, not provenance-only
L2	Fix actionable memory byte-identical; switch visible metadata	R5: target 10 tasks, capacity 5, 0 calls	Provenance-metadata effect if support gate passes
L3	Repeat L1/L2 in original model/runtime/source artifacts	Financial AgentDojo replication debt	Transport evidence only

Table 2: Provenance-conditioned ReasoningBank outcomes from the released financial-agent audit.

Retrieved memory provenance	Cases	Correct	Accuracy
Success-derived	101	94	0.931
Failure-derived	34	22	0.647

2.3 IDENTIFICATION STATEMENTS

Let $Y_W(w) = H(x', G(\tau, w), L_0, S)$ denote the potential outcome under writer mode w with metadata fixed at L_0 , and let $Y_L(\ell; m) = H(x', m, \ell, S)$ denote the potential outcome under metadata ℓ with actionable memory fixed exactly at m .

Proposition 1 (rung validity). (i) L0 does not identify either Δ_W or Δ_L without an ignorability assumption for source provenance, because a latent acquisition difficulty can jointly affect P and Y . (ii) Within a preregistered L1 admitted set, if writer mode is the only manipulated channel, future state E is fixed, and consistency/no-interference hold, the paired task-level contrast identifies the finite-set writer-mode effect $\mathbb{E}[Y_W(W_F) - Y_W(W_S)]$ for that admitted set. (iii) L1 does not identify Δ_L when $G(\tau, W_F) \neq G(\tau, W_S)$. (iv) Under L2, if $M = m$ is byte-identical and metadata is the only manipulated channel, the paired task-level contrast identifies $\mathbb{E}[Y_L(F; m) - Y_L(S; m)]$ for the evaluated substrate. L3 changes the target domain, not the intervention channel, so transport assumptions are additional to (ii)–(iv).

The non-upgrade in (iii) is constructive: two response functions can agree on every L1-observed point (M_S, L_0) and (M_F, L_0) yet differ arbitrarily on the unobserved metadata counterfactuals (m, L_S) and (m, L_F) . Thus no amount of L1 replication alone determines an L2 effect. For our L1 gate, malformed memories, answer leakage, insufficient embedding similarity, verifier-detected actionable asymmetry, or a unique task-solving fact are rejected before outcomes; this narrows the writer-mode bundle but does not supply the missing equality assumption. Consequently a strong L0 signal cannot be promoted to L1/L2, a non-decisive L1 result cannot rescue L2, and an L2 support failure is not counterevidence. The rung and independent unit are therefore part of the claim, not bookkeeping.

3 RELEASED EVIDENCE FOR A PROVENANCE EFFECT

We begin from the public aggregate audit of ReasoningBank on benign AgentDojo Banking endpoints (Li & Zhu, 2026). The audit separates retrievals according to whether the retrieved memory was derived from a successful episode or from a failed episode. Table 2 reproduces the reported counts after converting the rounded accuracies back to their integer support.

The observed accuracy gap is 28.4 percentage points. This is a large effect at the retrieval stratum level. The transition audit gives the provenance signal a sharper interpretation. ReasoningBank records 25 $W \rightarrow C$ recoveries and 9 $C \rightarrow W$ regressions. Ten $W \rightarrow W$ persistent failures remain after evolution. All ten belong to the failure-derived retrieval stratum.

The rounded provenance accuracies imply 12 failures among 34 failure-derived retrievals and 7 failures among 101 success-derived retrievals. The ten $W \rightarrow W$ cases therefore account for roughly 83.3% of failures in the failure-derived stratum. The success-derived stratum contains no $W \rightarrow W$ case in the audit. Figure 2 visualizes this decomposition.

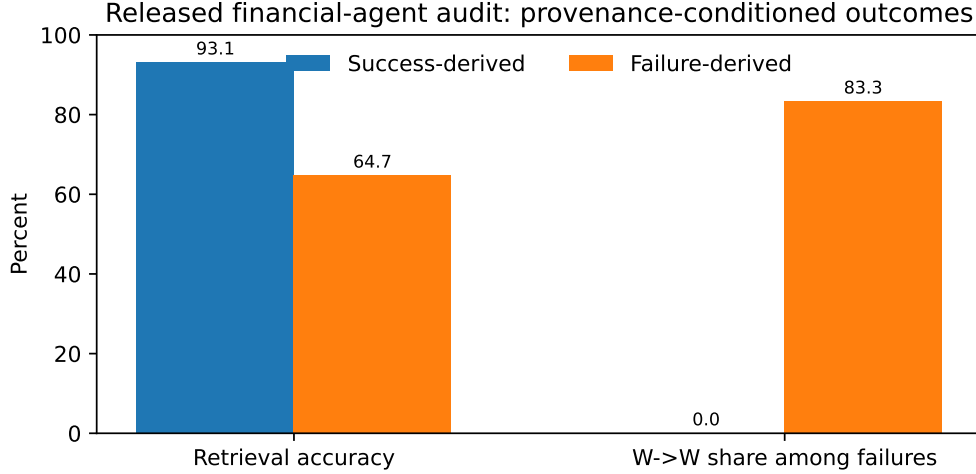


Figure 2: The released audit shows a large accuracy gap by memory provenance. Persistent $W \rightarrow W$ failures are concentrated in failure-derived retrievals.

3.1 THE SIGNAL IS STRONGEST FOR PERSISTENCE

The transition structure changes the scientific interpretation of the aggregate accuracy gap. Failure-derived memory is tightly associated with errors that persist across self-evolution. The source audit identifies task difficulty as a confound for this association. Our causal formulation preserves the persistence signal as the main target and removes task difficulty through an intervention on memory provenance.

A simple arithmetic check also constrains the mechanism. The failure-derived stratum contains roughly two failures beyond the ten $W \rightarrow W$ cases. The success-derived stratum contains seven failures. The provenance signal therefore concentrates on persistent error much more strongly than on newly induced regression. This pattern motivates a paper claim centered on failure persistence.

3.2 INDEPENDENT MECHANISM EVIDENCE

Memory Reward Inflation studies a different memory architecture and identifies an amplification channel for erroneous memories (Asadolahi et al., 2026). Incorrect memories can receive inflated utility scores, which increases their future reuse. The paper also shows that this behavior can persist under similarity-based retrieval. This mechanism is compatible with the provenance effect studied here. A failure-derived memory can carry corrective advice while remaining anchored to a trajectory whose process already demonstrated an error mode. Repeated reuse can then sustain that mode when the corrective abstraction is incomplete.

Together, the financial-agent audit and reward-inflation results motivate a causal model in which memory provenance can change the distribution of future behavior. Section 4 tests the directional provenance hypothesis with two frozen controlled endpoints.

4 CONTROLLED WRITER-MODE BRIDGE

We execute matched outcome-conditioned writer-mode interventions on released WebArena evidence using the first-party ReasoningBank memory-writing contract. The financial AgentDojo audit supplies the motivating association. Within each downstream pair we fix the future query and observation evidence and apply a preregistered information-equivalence gate to the generated memories. Because the success and failure branches use different reflection objectives, this bridge tests the writer-mode bundle associated with provenance rather than changing provenance metadata alone.

Table 3: L1 terminal-success writer-mode contrast. Δ is success-derived minus failure-derived terminal success.

Released task	Success-derived	Failure-derived	Δ
167	0.000	0.000	0.000
23	1.000	0.667	+0.333
387	0.000	0.000	0.000
388	0.500	0.167	+0.333
Mean	0.375	0.208	+0.167

4.1 TERMINAL-SUCCESS ENDPOINT

We freeze six candidate future tasks before memory writing or downstream evaluation. For each task, the writer receives paired acquisition traces under its success-reflection objective and its failure-reflection objective. The downstream query evidence remains fixed within each pair.

A pair enters terminal evaluation only after an information-equivalence gate. Both memories must parse as valid ReasoningBank items. Their embedding cosine must be at least 0.75. An independent verifier must judge the actionable guidance equivalent and find no task-solving fact unique to either memory. Literal benchmark-reference leakage excludes a pair before downstream outcomes. Four of the six frozen candidates pass. Their cosine values range from 0.902 to 0.931.

The evidence packets contain repeated global browser chrome. A condition-blind projection removes complete menu blocks before the clean run. It also drops blank lines and exact duplicate lines. The projection is frozen without consulting downstream answers. A context preflight shows that the largest prompt occupies 5,209 input tokens inside a 12,288-token allocation. Qwen2.5-32B remains the executor at temperature 0.6. The pair set and seed schedule remain frozen.

Each qualified pair receives six paired seeds per writer condition. Terminal correctness uses the WebArena must-include evaluator. Table 3 reports pair-level success rates.

The mean controlled difference is +0.1667. The preregistered support gate requires a difference of at least 0.15 together with a one-sided stratified permutation value below 0.05. The magnitude criterion is met. The 100,000-permutation test gives $p = 0.0785$, so the support gate remains closed. More fundamentally, four independent task units permit only $2^4 = 16$ exact sign patterns; the smallest attainable exact one-sided value is therefore $1/16 = 0.0625$. R4 is structurally incapable of clearing the frozen 0.05 gate and is treated as directional, non-decisive evidence. The counterevidence gate remains closed as well.

4.2 DISJOINT EARLY-ACTION ENDPOINT

The manuscript specifies action divergence before terminal feedback as a second endpoint. We execute that endpoint on a disjoint cohort after the terminal run. Candidate selection occurs before counterfactual memory writing. A released successful trajectory supplies a frozen reference action at an intermediate browser state. This reference is external to the L1 writer-mode contrast.

The candidate rule selects the earliest visible single-click decision at step two or later. Prior D2 task IDs are excluded. The remaining tasks are ordered by numeric task ID. The first 24 candidates are frozen. Their compact process traces are byte-identical across writer conditions. Terminal success labels are removed from those traces before the ReasoningBank success-reflection and failure-reflection interventions.

Twenty-three of the 24 candidates pass the pre-outcome information-equivalence gate. The minimum gate is 12. One candidate is excluded because the independent verifier returns an unparsable audit response; this support failure carries no scientific verdict. All 23 eligible pairs enter downstream execution. Each receives six paired seeds under each memory condition, producing 276 completed requests with zero support failures.

The endpoint asks for the next click from the frozen browser state. The released successful trajectory provides the reference click. We use this endpoint only to localize action-selection divergence under the L1 writer-mode contrast before terminal feedback. Reference-action match is not treated as a

Table 4: Two frozen L1 writer-mode endpoints. Support requires $\Delta \geq 0.15$ with $p_S < 0.05$. Counterevidence requires $\Delta \leq -0.15$ with $p_F < 0.05$.

Endpoint	Units	Requests	Δ	Relevant p	Gate
Terminal success	4	48	+0.167	$p_S = 0.0785$	closed
Reference-action match	23	276	-0.094	$p_F = 0.0792$	closed

surrogate for task success: a different click can still lead to a correct trajectory, so terminal quality remains the distinct endpoint above. Statistical inference uses tasks as the independent units. The preregistered test sign-flips each task-level success-derived minus failure-derived reference-match difference over 100,000 permutations.

For the early-action endpoint, the observed success-derived minus failure-derived reference-action match difference is -0.0942 . Its support-side value is 0.9841 . The counterevidence-side value is 0.0792 . Paired action choices differ in 21.0% of seeds. Error-consistent divergence in the hypothesized direction occurs in 0.7% of seeds. The reverse pattern occurs in 10.1%. These action-level quantities are mechanism-localization evidence only; they do not establish that either writer condition has higher terminal task quality.

4.3 CROSS-ENDPOINT ADJUDICATION

Table 4 shows opposite signs across two different readouts. The R4 terminal endpoint points toward the frozen C4 hypothesis, but its four-unit test is non-decisive under the frozen significance rule. R6 measures early action selection and does not adjudicate terminal task quality. Its negative reference-match difference therefore cannot be promoted to terminal counterevidence. Neither preregistered gate passes. We keep C4 as an active unrefuted hypothesis and assign no supported causal sign to the current controlled evidence. The frozen thresholds remain unchanged.

C5 remains a design implication. Provenance is an upstream variable in the memory writer, and the two controlled writer-mode conditions exhibit paired behavioral differences under matched query state. These experiments do not assign those differences a supported provenance-only causal sign, and a downstream performance benefit from provenance-aware governance remains outside the supported claim set.

5 RELATED WORK

Content- and utility-centric agent memory. A-MEM builds dynamically linked agentic memories (Xu et al., 2025), while MemRL learns runtime utility over episodic memories (Zhang et al., 2026). ReasoningBank, Agent Workflow Memory, and Reflexion likewise turn trajectories or feedback into persistent reusable state (Ouyang et al., 2026; Wang et al., 2024; Shinn et al., 2023). These systems establish memory construction and retrieval. Our audit asks which history-dependent channel is being manipulated when future behavior is attributed to provenance.

Provenance and trust. Execution-provenance and evidence-tracing work treats lineage and provenance-bearing memory as trust objects (Wang et al., 2026). Memory Provenance Laundering shows that consolidation can hide low-trust source provenance while preserving action triggers and proposes a provenance firewall (Xu et al., 2026). These works establish provenance as a governance object. Our closest-work boundary is narrower: success-derived versus failure-derived *process* provenance, with a distinction between an outcome-conditioned writer-mode intervention (L1) and a byte-identical provenance-metadata intervention (L2).

Learning from failure and memory quality. Failure-derived information can be beneficial. Learning from Failure converts failed computer-use trajectories into inference-time diagnoses and patches (Sun et al., 2026); Rethinking Self-Evolving Agent Skills reports useful evolved skills in feedback conditions containing failures (Liu et al., 2026). Memory Reward Inflation separately shows how erroneous memories can be overvalued and propagated (Asadolahi et al., 2026). Together these results

prevent a blanket “failure memory is bad” interpretation and motivate an audit with no assumed causal sign.

Causal identification and claim boundary. The financial-agent audit reports a strong success/failure provenance association (Li & Zhu, 2026), but task difficulty can confound that gap. Existing memory ablations can jointly change information, generation process, retrieval, and metadata. Our L0–L3 protocol requires the intervention channel, hold-fixed variables, independent inference unit, and strongest allowed claim to be declared separately. R4 and R6 are L1 worked examples: they hold future query/state fixed and use a preregistered operational semantic-equivalence rule, but generated memories need not be byte-identical. R5 is the cleaner L2 design because actionable memory is fixed exactly and only provenance metadata changes; its frozen cohort is support-insufficient and executes zero model calls. The contribution is therefore the identification object and resulting unresolved boundary, not a directional causal theorem or a validated provenance-governance policy.

6 LIMITATIONS

The financial AgentDojo audit remains the source of the 0.931-versus-0.647 provenance association and the ten persistent $W \rightarrow W$ cases. Its public release does not expose the per-query ReasoningBank artifacts required for an exact matched swap. A reconstruction preflight recovered the benchmark and ReasoningBank implementation, but the source configuration uses Qwen 3.7 Flash with qwen3.7-text-embedding and the current environment lacks auditable access to that configuration. Source-faithful L3 replication therefore remains support and experiment debt, not a scientific negative.

The L1 writer-mode evidence is intentionally bounded by its independent units. R4 contains four task pairs and 48 requests; its observed terminal difference is $+0.1667$ with $p = 0.0785$. With four units, the exact one-sided sign-flip test has only 16 patterns and cannot attain $p < 0.05$. R6 contains 23 disjoint task pairs and 276 requests; its observed reference-action difference is -0.0942 with counterevidence-side $p = 0.0792$. Reference-action match localizes action selection and is not a validated terminal-quality surrogate. Both frozen gates remain closed, and no threshold changes after observing either endpoint.

Operational semantic equivalence is weaker than exact information identity. R4/R6 require valid memory parsing, an embedding-cosine floor, an independent verifier, and no unique task-solving fact or literal reference leakage. These checks reduce obvious information asymmetry but cannot exclude every decision-relevant wording, emphasis, or decomposition difference introduced by the success-versus failure-reflection objectives. Consequently L1 identifies, at most, the declared writer-mode bundle. The stricter L2 design holds the actionable core byte-identical and changes only structured provenance metadata, but its frozen ten-task target exceeds the five unique tasks available under the preregistered released-shopping contract. R5 therefore stops with zero model calls and no scientific verdict; lowering the target afterward would violate the design.

The reported financial accuracies are rounded to three decimals. The integer correct counts used in our decomposition are the unique counts consistent with the released supports and rounded values, while transition allocation beyond the ten $W \rightarrow W$ cases is derived from public aggregates. More broadly, the controlled bridge uses WebArena while the motivating signal comes from financial AgentDojo. Provenance-aware governance has not been shown to improve downstream performance. These boundaries keep U1 and U2 unsupported.

7 CONCLUSION

The paper’s primary methodological output is a provenance-audit identification ladder. L0 asks whether future outcomes differ across observed provenance strata. L1 changes an outcome-conditioned writer under fixed future state and an operational equivalence rule. L2 holds actionable memory byte-identical and changes only visible provenance metadata. L3 tests whether an identified contrast transports to the motivating source runtime. The ladder makes the intervention channel, independent inference unit, identifying assumptions, and strongest allowed claim explicit.

Our evidence populates these rungs without manufacturing a causal sign. The released financial audit provides a strong L0 signal: 0.931 versus 0.647 accuracy and all ten reported persistent $W \rightarrow W$ failures

in the failure-derived stratum. R4 and R6 instantiate L1 but leave both frozen causal gates closed; R4 is structurally non-decisive with four units, while R6 uses a non-terminal action-localization endpoint. R5 targets L2 but support-stops at five eligible unique tasks for a frozen target of ten. Source-faithful L3 replication remains debt. C4 therefore remains active and unrefuted rather than supported or rejected.

This changes what a memory audit should report. A retrieved text is not a complete audit object when the process that produced it is discarded, but provenance is also not a single causal treatment until its channel is specified. Reporting the ladder rung prevents observational association, writer-mode effects, provenance-metadata effects, and transport evidence from being collapsed into one ambiguous “provenance effect.” Any policy that converts provenance into trust or reuse decisions still requires separate downstream validation.

REPRODUCIBILITY STATEMENT

The appendix records the frozen intervention rules and statistical gates, together with the support-failure boundaries used by the controlled experiments. The submission supplement contains the claim ledger and its content-addressed evidence manifest. It also packages experiment receipts with the source needed to rebuild the manuscript and recompute the reported aggregate statistics.

AI USE DISCLOSURE

LLM-based tools assisted literature triage and earlier hypothesis formulation. They supported experimental-methodology design under frozen claim contracts and helped implement experiment code. They also assisted result interpretation and manuscript editing. Separate LLM reviewers were used for mock-review analysis. Scientific claims and reported numerical results were checked against frozen experiment artifacts and deterministic evaluation scripts; AI-generated judgments were not used as task-success ground truth. The authors reviewed the AI-assisted work and take responsibility for the final content.

REFERENCES

- Mohammad Asadolahi, Amir Amini, Samira Talebi, Amirfarhad Farhadi, and Azadeh Zamanifar. Memory reward inflation in self-improving llm agents, 2026.
- Jialong Li and Jialing Zhu. Auditing self-evolution in financial agents: Capability gains, security drift, and execution-interface mismatch, 2026.
- Yuxuan Liu, Zhaochen Su, Yuhao Zhang, et al. Rethinking self-evolving agent skills: Feedback dynamics over multiple rounds, 2026.
- Siru Ouyang, Jun Yan, I-Hung Hsu, Yanfei Chen, Ke Jiang, Zifeng Wang, Rujun Han, Long T. Le, Samira Daruki, Xiangru Tang, Vishy Tirumalashetty, George Lee, Mahsan Rofouei, Hangfei Lin, Jiawei Han, Chen-Yu Lee, and Tomas Pfister. Reasoningbank: Scaling agent self-evolving with reasoning memory. In *International Conference on Learning Representations*, 2026.
- Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems*, 2023.
- Xueqiao Sun, Xiaohan Wang, Ludwig Schmidt, et al. Learning from failure: Inference-time self-improvement for computer-use agents, 2026.
- Yiqi Wang, Jiaqi Zhang, Taotao Cai, et al. From agent traces to trust: A survey of evidence tracing and execution provenance in llm agents, 2026.
- Zora Zhiruo Wang, Jiayuan Mao, Daniel Fried, and Graham Neubig. Agent workflow memory, 2024.
- Jinghan Xu, Yiyong Xiao, Wanru Shao, et al. Memory provenance laundering in llm agents: A non-amplification firewall for persistent memory, 2026.

Wujiang Xu, Zujie Liang, Kai Mei, et al. A-mem: Agentic memory for llm agents, 2025.

Shengtao Zhang, Jiaqian Wang, Ruiwen Zhou, et al. Memrl: Self-evolving agents via runtime reinforcement learning on episodic memory, 2026.

A SOURCE-GROUNDED REANALYSIS AND INTERVENTION DETAILS

A.1 COUNT RECONSTRUCTION

The financial audit reports 101 success-derived retrievals at 0.931 accuracy and 34 failure-derived retrievals at 0.647. The unique integer numerators consistent with three-decimal rounding are 94 and 22, yielding 7 and 12 failures. It also reports ten $W \rightarrow W$ persistent failures, all failure-derived; hence 10/12 failure-derived errors are persistent, while the success-derived stratum contains no $W \rightarrow W$ case.

A.2 TERMINAL INFORMATION-EQUIVALENCE CONTRACT

Six terminal candidates are frozen before outcomes. Both memories use the same writer model; eligibility requires valid ReasoningBank items, paired embedding cosine at least 0.75, an independent verifier judging actionable guidance equivalent with no unique task-solving fact, and no literal benchmark-reference leakage. Four candidates pass: tasks {387, 167, 23, 388} with cosines 0.931, 0.902, 0.906, and 0.904. Task 385 is excluded for repeating “Cally”; task 164 repeats “Dry” and “Uneven color.” All exclusions are pre-outcome.

A.3 TERMINAL RUNTIME REPAIR BOUNDARY

A condition-blind projection removes repeated global menu blocks, blank lines, and exact duplicate lines without consulting references or outcomes. An initial context-allocation support failure is discarded before scientific admission; the clean restart changes only context capacity. Its largest prompt uses 5,209 of 12,288 input tokens; executor, temperature, seed schedule, pair set, scorer, effect floor, permutation seed, and significance rule remain frozen.

A.4 TERMINAL STATISTICAL RULE

Each qualified pair receives six paired seeds. The statistic is task-level terminal success, success-derived minus failure-derived. Support requires $\Delta \geq 0.15$ and one-sided $p < 0.05$; counterevidence requires $\Delta \leq -0.15$ and its one-sided $p < 0.05$, using 100,000 frozen-seed permutations. Observed $\Delta = +0.1667$ with $p_S = 0.0785$. Four independent task units imply only 16 exact sign assignments and a minimum one-sided value $1/16 = 0.0625$, so this endpoint cannot satisfy the frozen 0.05 rule.

A.5 EARLY-ACTION COHORT AND L1 WRITER-MODE CONTRAST

The disjoint WebArena cohort excludes prior D2 task IDs. A candidate must come from a released successful trajectory and expose a visible single-click reference action at step two or later; eligible tasks are sorted by ID and the first 24 are frozen before counterfactual writing. A compact process trace preserves instruction and action sequence but removes terminal success. The same trace is passed through first-party ReasoningBank success- and failure-reflection prompts using Qwen2.5-32B at temperature zero, directly varying outcome-conditioned writer mode while holding the trace fixed.

Eligibility again requires valid memory items, cosine at least 0.75, verifier-equivalent actionable guidance, no unique task-solving fact, and no reference-action leakage. Twenty-three pairs pass; candidate 159 is excluded for an unparsable verifier output. All 23 enter downstream execution.

A.6 EARLY-ACTION STATISTICAL RULE

Each task receives six paired seeds per writer condition with identical future query and frozen browser state. The output is one visible click index; the released successful trajectory supplies the reference. Inference is task-level: the permutation test sign-flips each task’s success-derived minus failure-derived reference-match difference over 100,000 draws, with the same ± 0.15 magnitude and

one-sided 0.05 gates. The 23 tasks yield 276 complete requests and zero support failures; mean difference is -0.0942 , with $p_S = 0.9841$ and $p_F = 0.0792$. Paired actions differ in 21.0% of seeds; hypothesis-direction error-consistent divergence is 0.7%, versus 10.1% in reverse.

A.7 EXACT-INFORMATION SUPPORT-CAPACITY STOP AND REOPEN CONDITION

R5 holds the actionable memory core byte-identical and changes only structured provenance metadata. Its frozen contract requires ten unique review tasks, but the released shopping split contains only five; it therefore stops before any model call, with no scientific verdict and no post-audit target reduction. The strongest reopen path is source-faithful financial AgentDojo replication of an L1 writer-mode or L2 exact-information contrast when query-level artifacts or auditable exact-model access become available. Such replication addresses L3 transport, not identification by itself; any new bridge must be frozen before outcomes and cannot reuse R4/R6 for threshold rescue.